



SE3082 - Parallel Computing

Lab-07

IT24101503

A. A. B. Shamim Basha

Exercise 07:

- **sum_bcast**
 - **Collective:** MPI_Bcast
 - **Array allocation:** Full array on every process
 - **Root manual sum:** Yes, using **MPI_Recv**
 - **Result location:** Root only
- **sum_scatter**
 - **Collectives:** MPI_Scatter
 - **Array allocation:** Full array on root; chunk on other processes
 - **Root manual sum:** Yes, using **MPI_Recv**
 - **Result location:** Root only
- **sum_gather**
 - **Collectives:** MPI_Scatter, MPI_Gather
 - **Array allocation:** Full array on root; chunk on other processes
 - **Root manual sum:** Yes, sums gathered values
 - **Result location:** Root only
- **sum_reduce**
 - **Collectives:** MPI_Scatter, MPI_Reduce
 - **Array allocation:** Full array on root; chunk on other processes
 - **Root manual sum:** No
 - **Result location:** Root only
- **sum_allreduce**

- **Collectives:** MPI_Scatter, MPI_Allreduce
- **Array allocation:** Full array on root; chunk on other processes
- **Root manual sum:** No
- **Result location:** Every process

- **sum_scan**

- **Collectives:** MPI_Scatter, MPI_Scan
- **Array allocation:** Full array on root; chunk on other processes
- **Root manual sum:** No
- **Result location:** Each process receives a different prefix sum; the last rank receives the total

Thinking Question

- Choose MPI_Scan when every process needs the cumulative result up to its own rank.
processes can use MPI_Scan on their record counts to calculate globally unique offsets when writing records into one output array.